Aggregate Nulls, Compositional Shifts: AI Exposure and the Junior Employment Share in UK Occupations

Extended Essay

EC423 — Labour Economics

Candidate Number: XXXXX

Word Count: XXXX

Abstract

Aggregate studies of AI's labour market effects consistently report small or null employment impacts at the occupation level. I argue that these results mask offsetting compositional shifts within occupations: AI displaces junior workers whose tasks overlap with AI capabilities, while the model predicts higher marginal products for senior workers whose expertise-intensive tasks are complementary. I formalise this mechanism by extending the expertise framework of Autor and Thompson (2025) to permit within-occupation task specialisation in a partial-equilibrium CES model. Using individual-level UK position data from Revelio Labs matched with LLM exposure scores, I exploit the unanticipated release of ChatGPT in November 2022 as a natural experiment. Occupations with higher AI exposure experience a monotonically increasing decline in the junior hiring share, reaching 11.5 percentage points by 2025. Senior workers in exposed occupations become substantially more mobile, a pattern that indicates increased senior reallocation but cannot distinguish senior augmentation from exit from deteriorating matches, while junior workers experience reduced transition opportunities. Conditional on transitioning, juniors in exposed occupations achieve larger seniority gains — an effect whose point estimate is stable after Heckman correction, suggesting genuine skill-upgrading.

1 Introduction

Since the launch of ChatGPT in autumn 2022, generative AI has undergone rapid development over the past three years. Beyond text generation, AI has achieved notable breakthroughs across a range of creative domains — including code generation, image synthesis, music composition, and video production. The pace of these advances has far exceeded the expectations of most observers, raising pressing questions about how AI will affect the labour market.

The task framework of Acemoglu & Restrepo (2019) predicts that automation can create new jobs while simultaneously displacing existing tasks. A central implication is that the net effect on aggregate employment is theoretically ambiguous, depending on the relative strength of these competing forces. Earlier waves of computerisation did produce measurable structural change: Autor & Dorn (2013) document significant occupational polarization in the United States as routine-cognitive and routine-manual tasks were automated, hollowing out middle-skill employment while expanding both tails of the skill distribution. One might therefore expect the strong diffusion of large language models and other generative AI systems since late 2022 to generate similarly visible labour market disruption. Yet the emerging post-ChatGPT evidence remains more muted at conventional levels of aggregation than this technological narrative would suggest. Massenkoff & McCrory (2026) find no systematic increase in unemployment among highly exposed occupations since late 2022, while Brynjolfsson et al. (2025) document sharper employment losses among early-career workers in AI-exposed occupations rather than a broad employment collapse. This pattern appears puzzling given the breadth of tasks that generative AI can now perform. Eloundou et al. (2024) estimate that roughly 80 percent of the US workforce holds occupations in which at least 10 percent of tasks are exposed to large language models, yet the expected

employment disruption has not materialized at conventional levels of aggregation. There are, however, early indications that the apparent absence of effects may itself be an artefact of aggregation. Lichtinger & Hosseini Maasoum (2025), using US résumé and job posting data, find that generative AI disproportionately affects workers by seniority level, a pattern they term *seniority-biased technological change*. Their findings suggest that the locus of AI's labour market impact lies not between occupations but *within* them—a possibility that aggregate studies are ill-equipped to detect, which is also the biggest source of this essay's motivation.

I argue that these aggregate null results are not evidence of absence, but rather reflect offsetting compositional shifts within occupations. The central thesis of this essay is that AI displaces junior workers, whose task portfolios overlap heavily with the capabilities of current AI systems, while simultaneously augmenting senior workers, whose expertise-intensive tasks are complementary to AI. Because both groups are classified under the same occupation codes, the displacement of juniors and the entrenchment of seniors partially cancel when measured at the occupation level, producing the observed near-zero aggregate effects. This prediction aligns with a growing body of anecdotal and empirical evidence that firms have sharply curtailed entry-level hiring while increasing demand for — and compensation of — experienced workers (Brynjolfsson et al., 2025).

I extend the expertise framework of Autor & Thompson (2025) by relaxing their assumption of within-occupation worker homogeneity. In the resulting partial-equilibrium CES model, junior and senior workers possess comparative advantage in different task types, and AI enters as a factor input that substitutes for junior labour in low-expertise tasks while complementing senior labour through the CES production structure. The model yields four formal predictions: 1. a declining junior employment share in exposed occupations; 2. rising demand for senior labour through complementarity; 3. a near-zero net effect at the occupation level; 4. wage movements accompanying the quantity response, with junior wages falling and senior wages rising.

Measuring the impact of AI is hard. Unlike discrete, large-scale shocks such as the COVID-19 pandemic, constructing a credible counterfactual for the impact of AI on the labour market is particularly challenging, precisely because AI adoption is gradual and diffuse rather than sudden (Massenkoff & McCrory, 2026). I test the model's direct compositional and wage implications using an individual-level panel of UK workers constructed from Revelio Labs, matched with the AI occupational exposure scores of Eloundou et al. (2024). This dataset also allows me to examine career trajectories, seniority transitions, and occupation switches before and after the release of ChatGPT in November 2022, exploiting cross-occupation variation in AI exposure intensity. I document a significant decline in the junior share of positions within highly exposed occupations following the AI shock, consistent with displacement. I then study mobility and seniority gains as complementary adjustment margins that are motivated by the model's logic but not formally derived from its static structure. Conditional on transitioning, junior workers in exposed occupations experience larger seniority gains — an effect whose point estimate is stable after Heckman selection correction, suggesting genuine skill-upgrading. Senior workers in exposed occupations become substantially more mobile, suggesting increased senior reallocation. Because this estimate captures gross transitions rather than net headcount changes, however, it does not by itself distinguish senior augmentation from exit from deteriorating matches. In the Revelio predicted salary data, I do not detect the wage movements predicted by the model, and I find no evidence of occupation switching in response to AI exposure. These findings fill an empirical gap explicitly identified by Brynjolfsson et al. (2025) and Klein Teeselink (2025), who call for individual-level trajectory analysis to complement their aggregate and firm-level results.

The remainder of this essay proceeds as follows. Section 2 develops the theoretical framework. Section 3 describes the data and identification strategy. Section 4 presents

the empirical results on compositional shifts and career mobility. Section 5 discusses limitations and situates the findings within the broader literature. Section 6 concludes.

2 Theoretical Framework

Autor & Thompson (2025) introduce an "expertise" framework in which the effect of task automation depends not only on the number of tasks displaced, but also on the *expertise level* of those tasks relative to the remaining task bundle. When automation replaces *inexpert* tasks, the expertise requirement rises; wages increase, while employment falls because fewer workers clear the higher threshold. When automation replaces *expert* tasks, the outcome reverses. This "expertise bifurcation" helps explain why routine task-intensive occupations have experienced declining employment but not uniformly declining wages.

Central to their framework is Assumption 3 (occupational task bundling): *each worker in an occupation must perform* all *of its non-automated tasks, and conditional on meeting the expertise threshold, all workers are equally productive. This inseparability property is well suited to their between-occupation analysis, where the relevant variation lies across occupations with differing task compositions.* However, it rules out within-occupation heterogeneity — precisely the margin this essay investigates.

I keep the core insight of Autor & Thompson (2025) that occupations bundle tasks of varying expertise levels, but relax the inseparability requirement. The structural motivation for this relaxation comes from models of knowledge hierarchies. Garicano & Rossi-Hansberg (2006) show that workers with heterogeneous skills are assigned to different roles within organisations: less experienced workers handle more routine problems, while more experienced workers solve harder problems and leverage their knowledge through teams. This provides a production-side basis for within-occupation task specialisation. I therefore allow junior and senior workers to allocate effort dispro-

portionately to tasks matching their comparative advantage. The Revelio Labs seniority measure is then used as an empirical proxy for this theoretically motivated distinction.

2.1 Model Setup

Consider a single occupation in partial equilibrium. Output Y is produced by combining three task types via CES aggregation:

$$Y = \left[\beta_L y_L^{\frac{\sigma-1}{\sigma}} + \beta_H y_H^{\frac{\sigma-1}{\sigma}} + \beta_G y_G^{\frac{\sigma-1}{\sigma}}\right]^{\frac{\sigma}{\sigma-1}} \tag{1}$$

where $\sigma > 0$ is the within-occupation task substitution elasticity and $\beta_L, \beta_H, \beta_G > 0$ are task share parameters summing to unity. For normalization, we let $\beta_L + \beta_H + \beta_G = 1$. The parameter σ governs how readily the occupation substitutes between task types; it is conceptually distinct from the between-occupation elasticity λ in Autor & Thompson (2025).

Following Autor & Thompson (2025), tasks fall into three categories:

- Task L: low-expertise activities, including routine operations that are readily automatable by AI.
- Task H: high-expertise activities, including context-dependent judgement, managerial decision-making, and other functions requiring accumulated experience.
- Task G: generic support tasks assigned to junior workers, including common execution-oriented activities that remain non-automated.

The inclusion of a junior-assigned generic task follows the framework of Autor & Thompson (2025), and is motivated by a distinctive feature of the current AI wave: unlike the industrial automation of the twentieth century, which primarily displaced manual and blue-collar labour, generative AI disproportionately affects white-collar occupations — precisely those traditionally regarded as requiring substantial investment in education (Webb, 2020). The impact of AI on occupational task structures thus operates in a top-down manner, penetrating first into the cognitive and professional tasks that define high-skill roles.

Each task type is produced as follows:

$$y_L = A \cdot M + a_L \cdot n_J^L, \quad y_H = b_H \cdot n_S, \quad y_G = c \cdot n_J^G \tag{2}$$

In task L, AI capital M (rented at rate r) and junior labour n_J^L are perfect substitutes, with A denoting AI capability. Task H requires senior workers exclusively: the judgement-intensive nature of these tasks imposes an expertise threshold that junior workers cannot clear, consistent with hierarchical expertise in Autor & Thompson (2025). Task G is modelled as a junior-assigned generic task; I abstract from senior production of G and treat senior labour as specialised in task H, reflecting the higher opportunity cost of senior time.[1]

Junior workers allocate their labour between tasks L and G subject to $n_J^L + n_J^G = n_J$. At the occupation level, labour supply responds to wages with constant elasticity:

$$n_J = \varphi_J w_J^{\varepsilon}, \quad n_S = \varphi_S w_S^{\varepsilon} \tag{3}$$

where $\varepsilon > 0$ captures workers' ability to enter or exit the occupation in response to wage changes.

2.2 Equilibrium and Comparative Statics

The competitive firm's cost minimisation yields three equilibrium conditions. The first-order conditions are derived in Appendix A.1, and the comparative statics are derived in Appendix A.3.

No-arbitrage in task L. Since AI and junior labour are perfect substitutes, the firm employs both only when their unit costs are equalised:

$$w_J = \frac{a_L \cdot r}{A} \tag{4}$$

Junior wages are pinned entirely by the cost-effectiveness of AI. As AI capability A rises, w_J falls — regardless of conditions in tasks H or G.

Internal allocation of junior labour. Equalizing the marginal revenue product of junior labour across tasks L and G yields a constant task output ratio $y_G/y_L = \gamma$, where

[1] This restriction is equivalent to assuming that the senior return in task H exceeds the return seniors would obtain from generic support work. See Appendix A.2.

$\gamma \equiv (\beta_G c/\beta_L a_L)^\sigma$ is a structural parameter independent of A. Although AI transforms the production of task L internally, the ratio of generic to low-expertise output is invariant to AI capability because both tasks share the same marginal factor at wage w_J.

Senior augmentation via complementarity. Senior wages satisfy:

$$w_S = p \cdot \beta_H \left(\frac{Y}{y_H}\right)^{\frac{1}{\sigma}} \cdot b_H \tag{5}$$

The term $(Y/y_H)^{1/\sigma}$ embodies the key channel. On impact, holding senior labour fixed, AI raises y_L and hence total output Y, while y_H remains unchanged. This raises the marginal revenue product of task H — and therefore w_S. In equilibrium, the induced increase in senior labour raises $y_H = b_H n_S$, which dampens but does not reverse the augmentation effect because the senior labour demand curve shifts outward. This is the within-occupation analogue of the between-occupation augmentation result in Autor & Thompson (2025).

Comparative statics. An increase in AI capability A produces the following adjustment. First, w_J falls via Equation 4, reducing n_J through elastic labour supply as junior workers exit to other occupations. Within the occupation, junior workers partially reallocate from task L to task G, but total junior employment nonetheless declines. Second, the rise in y_L shifts senior labour demand outward through CES complementarity. Senior wages w_S rise and, through elastic supply, n_S increases; the resulting increase in y_H partially offsets the initial rise in Y/y_H but leaves the equilibrium senior wage and employment above their initial levels. The combined effect — falling n_J and rising n_S — produces a decline in the junior employment share. The net effect on total occupation-level employment is theoretically ambiguous, depending on the magnitudes of σ and ε.

2.3 Testable Predictions

The static model formally delivers four implications:

1. **Compositional shift:** In occupations with higher AI exposure, the junior employment share declines following the AI shock, as junior workers are displaced from task L and exit the occupation.
2. **Senior augmentation:** Senior workers in high-exposure occupations experience rising marginal products, wages, and labour demand through CES complementarity.
3. **Wage adjustment:** Junior wages decline and senior wages rise. If the occupation-level supply elasticity ε is large, wage changes may be small relative to employment changes, but the model still predicts price adjustment.
4. **Aggregate ambiguity:** The net occupation-level employment effect is indeterminate in sign, consistent with the aggregate null results documented in the existing literature.

The generic task channel provides an additional nuance: because junior workers retain a productive role in task G, displacement from high-AI occupations is *partial* rather than total. The junior employment share declines but does not vanish, consistent with the patterns documented in the empirical analysis that follows.

The model is deliberately static and occupation-specific. It does not contain search, matching, multi-occupation mobility, or skill accumulation. The career-mobility analysis below should therefore be read as a complementary empirical investigation of worker-level adjustment margins suggested by the model, rather than as a direct test of additional formal predictions.

3 Data and Identification Strategy

3.1 Data and Specification

The data I use is the individual-level position data from Revelio Labs, which aggregates information from LinkedIn profiles in the United Kingdom. The sample comprises 23,305,509 distinct workers holding 50,742,348 position records that were

active at some point between January 2019 and December 2025, which spans before and after the release of ChatGPT in November 2022. For each position, I observe the O*NET occupation code, start and end dates, predicted salary data, company id, and two variables central to my analysis: an ordinal seniority measure ranging from 1 (Entry) to 7 (Senior Executive), and a within-individual position number that records the chronological order of each role. A detailed mapping of the seniority classification is provided in Table 1.

Seniority	Level	Examples
1	Entry	Accounting Intern, Software Engineer Trainee, Paralegal
2	Junior	Account Receivable Bookkeeper, Junior Software QA Engineer, Legal Adviser
3	Associate	Senior Tax Accountant, Lead Electrical Engineer, Attorney
4	Manager	Account Manager, Superintendent Engineer, Lead Lawyer
5	Director	Chief of Accountants, VP Network Engineering, Head of Legal
6	Executive	Managing Director (Treasury), Director of Engineering (Backend Systems), Attorney (Partner)
7	Senior Executive	CFO, COO, CEO

Table 1: Revelio Labs seniority classification

In practice, I define a worker as *junior* if their position carries a seniority level of 1 or 2, corresponding to Entry and Junior roles in Table 1.[2] These workers typically perform execution-oriented tasks under supervision. From the Associate level onward, workers begin to have some discretionary space rather than merely executing tasks assigned by superiors. In the language of the theoretical framework, junior workers have not yet cleared the expertise threshold required for task H. This classification captures 58.9 per cent of the sample. Lichtinger & Hosseini Maasoum (2025) validate this seniority measure by showing that the distribution of job-title keywords aligns with expected seniority distinctions and remains stable before and after the diffusion of generative AI.

To measure occupation-level exposure to AI, I use the LLM exposure scores constructed by Eloundou et al. (2024), who employ a combination of human annotators and

[2]The methodology of how Revelio Labs classify seniority is available at: https://www.data-dictionary.reveliolabs.com/methodology.html#seniority.

GPT-4 to assess, for each O*NET task, the share of task time that could be reduced by at least 50 per cent with access to a large language model. The resulting score enters all specifications as a continuous variable. I merge these scores to the Revelio positions data via O*NET–SOC codes, achieving an 89.5 per cent match rate; unmatched positions are excluded from the analysis. For robustness checks, I replicate the main specifications using two alternative measures: the AI Occupational Exposure index of Felten et al. (2021) and the AI Applicability Score of Tomlinson et al. (2025), with results reported in Appendix B.

The primary empirical specification is an event study that tracks the junior hiring share across occupations with differing AI exposure over time. To align the estimation window with the sample construction, the main compositional-shift analysis uses position starts from 2019H1 to 2025H2. I collapse the position-level data into occupation × half-year cells and estimate

$$\text{JuniorShare}_{o,t} = \sum_{k \neq k_0} \beta_k \big(m_{1,o} \times \mathbb{1}[t = k] \big) + \alpha_o + \gamma_t + \varepsilon_{o,t} \tag{6}$$

where $\text{JuniorShare}_{o,t}$ is the proportion of new hires classified as junior (seniority ≤ 2) in O*NET occupation o during half-year t; $m_{1,o}$ is the continuous LLM exposure score; α_o and γ_t are occupation and half-year fixed effects; and k_0 denotes the omitted baseline period, 2022H2. Although ChatGPT was released on 30 November 2022, I treat 2022H2 as a pre-adjustment reference period because measurable hiring responses are unlikely to appear immediately in position start dates. The coefficients β_k trace out the differential evolution of the junior share for a one-unit increase in AI exposure relative to this baseline period. I weight each cell by the number of underlying positions to account for heterogeneous cell sizes and cluster standard errors at the O*NET occupation level to permit arbitrary within-occupation serial correlation. As a complementary specification, I also report a pooled difference-in-differences estimate that replaces the period-specific interactions with a single $m_{1,o} \times \text{Post}_t$ term, where Post_t equals one from 2023H1 onward. An extended-window version that begins in 2018H1 is reported as a robustness

check in Appendix B, since starts before 2019 may be affected by left-censoring from the position-sample construction.

To examine career mobility, I further construct a worker-level transition dataset from the original positions dataset. A transition is defined as a consecutive pair of positions held by the same individual, with the transition date set to the start date of the subsequent position. I impose sample restrictions to ensure clean identification of job transitions, yielding 12.6 million valid transitions.[3] I define the post-treatment period for mobility as beginning in November 2022, corresponding to the month of ChatGPT's public release.

The extensive margin specification estimates a linear probability model of the form

$$\text{Transition}_i = \delta_1 \text{AI}_o \times \text{Post}_i + \delta_2 \text{Junior}_i \times \text{AI}_o \times \text{Post}_i + \boldsymbol{X}_{i'}\eta + \alpha_o + \gamma_{s(i)} + \varepsilon_i \quad (7)$$

where Transition_i equals one if origin-position observation i is followed by a move to a different position within the panel window; AI_o is the continuous LLM exposure score of the origin occupation; Junior_i equals one if the origin position has seniority ≤ 2; and $\boldsymbol{X}_i$ includes the lower-order interactions. The function $s(i)$ denotes the start month of the origin position, which indexes the month fixed effects. The post indicator Post_i is based on the observed transition date when a transition occurs and on the origin start date otherwise. The coefficient δ_1 captures the differential change in transition probability for senior workers in high-exposure occupations after ChatGPT, while δ_2 captures the additional differential for junior relative to senior workers. I estimate the intensive margin—conditional on a transition occurring—using the change in seniority level as the outcome:

$$\Delta\text{Seniority}_i = \theta_1 \text{AI}_o \times \text{Post}_i + \theta_2 \text{Junior}_i \times \text{AI}_o \times \text{Post}_i + \boldsymbol{X}_{i'}\mu + \alpha_o + \gamma_{\tau(i)} + u_i \quad (8)$$

where $\Delta\text{Seniority}_i$ is the difference in seniority between the destination and origin positions, and $\tau(i)$ is the transition month. The coefficient θ_2 captures whether junior

[3] I exclude individuals with more than 50 recorded positions (likely data artefacts), drop all positions for individuals with any overlapping employment spells to avoid ambiguous transition timing, and remove positions with missing start dates.

workers in high-exposure occupations experience differentially larger seniority gains upon transition in the post period.

The intensive-margin estimates condition on a transition having occurred, which raises a sample selection concern (Heckman, 1979). If the composition of the transition sample changes systematically after ChatGPT — for example, if weaker junior workers are disproportionately displaced, leaving only stronger workers to transition — the seniority gains in Equation 8 may be biased. I address this with a Heckman two-step procedure. The selection equation models the probability of transitioning using the main regressors, occupation fixed effects, and a firm-level exclusion restriction: the pre-period mean separation rate at the worker's origin firm, entered both as a main effect and interacted with junior status. The logic is that firms with historically higher turnover generate more transitions mechanically, but conditional on transitioning, the origin firm's historical separation rate should not affect the seniority change achieved in the new position. The inverse Mills ratio from the selection equation is then included in the outcome equation to correct for non-random selection into the transition sample. I do not implement a Murphy–Topel or bootstrap correction for the second-stage standard errors, so the standard errors in the corrected columns should be interpreted as potentially understated rather than fully corrected.

3.2 Identification Strategy

The causal interpretation of the event-study coefficients rests on two conditions: that occupation-level AI exposure is not itself shaped by post-treatment labour market adjustment, and that the timing of the shock was unanticipated. The first condition holds by construction. The LLM exposure scores of Eloundou et al. (2024) are derived from O*NET task descriptions that predate ChatGPT; they capture how susceptible an occupation's task bundle is to large language models, not whether firms have actually adopted AI. This is an important distinction from Lichtinger & Hosseini Maasoum

(2025), who rely on firm-level variation in AI adoption — a measure that may be endogenous to workforce composition decisions. In my setting, exposure is pinned down by the pre-existing task content of jobs and does not respond to post-shock employer behaviour.

The second condition is supported by the sudden public arrival of ChatGPT. Figure 1 shows that the Google Trends index for "ChatGPT" was zero before its release on 30 November 2022 and rose sharply thereafter, confirming that the technology became publicly salient only at the end of 2022H2. This timing supports the use of 2022H2 as the omitted period in Equation 6. Although the release occurred within that half-year, the remaining window in 2022 was too short for hiring adjustments to appear in position start dates. Moreover, Lichtinger & Hosseini Maasoum (2025) show that firm-level hiring responses began in the first quarter of 2023. I therefore define the post period for the compositional-shift analysis from 2023H1 onward and use 2022H2 as the last pre-adjustment baseline.

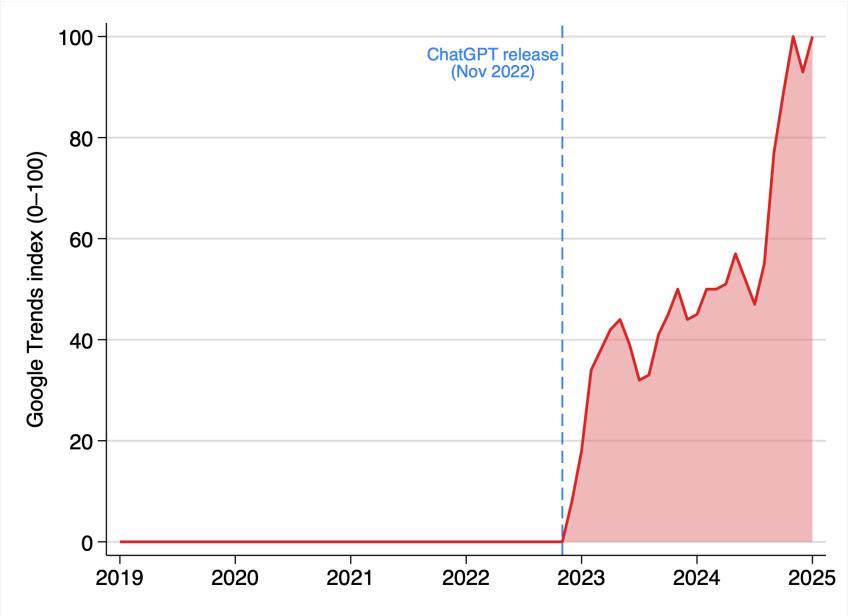

Figure 1: Google Trends index for the keyword "ChatGPT", 2019–2025

The key identifying assumption behind Equation 6 is that, absent generative AI, the junior hiring share would have evolved similarly across occupations with different levels of LLM exposure. This assumption cannot be tested directly, but the event-study design

offers a useful diagnostic: if the pre-treatment coefficients $\hat{\beta}_k$ are close to zero and show no systematic trend, it is harder to argue that the post-period divergence reflects pre-existing dynamics rather than the AI shock. I report this assessment in Section 4, together with a formal linear pre-trend test following Freyaldenhoven et al. (2019).

Two limitations of this design deserve mention. First, because AI exposure is measured at the occupation level, I cannot include firm × time fixed effects of the kind used by Lichtinger & Hosseini Maasoum (2025). My estimates therefore cannot rule out the possibility that firm-level confounders correlated with occupation-level exposure account for part of the observed compositional shift. Second, the seniority threshold that defines junior status is treated as fixed, yet AI may itself change the expertise requirements within occupations over time, shifting the boundary between junior and senior work endogenously. I hold this boundary constant throughout the analysis and note that modelling its endogenous response would require the between-occupation general equilibrium framework of Autor & Thompson (2025), which is beyond the scope of this essay.

4 Results

I begin by establishing the central compositional fact: the junior share of new hires in high-AI-exposure occupations declined significantly after ChatGPT. This result is related to the firm-level evidence in Klein Teeselink (2025), who documents weaker outcomes for junior workers in AI-exposed firms, but the two estimates are not directly comparable. My outcome is a hiring share within occupation–half-year cells, whereas that study examines employment levels at the firm level. The estimates therefore speak to different margins while pointing to the same broad pattern of junior workers bearing the early adjustment. The contribution of this section is twofold. First, I provide a more rigorous identification check through event-study analysis with occupation-specific

trend controls. Second, I lay the empirical foundation for the career mobility analysis that follows.

4.1 Compositional Shift

The present analysis focuses on a finer occupational margin: individual position starts classified by O*NET code. It also subjects the compositional shift to a more demanding identification strategy through event-study estimation.

Figure 2 plots the monthly junior share of new position starts, separately for occupations above and below the median AI exposure score. Prior to late 2022, both series fluctuate around stable means: approximately 55 per cent for the high-exposure group and 85 per cent for the low-exposure group. The higher baseline share of junior workers in low-exposure occupations is unsurprising, as these roles tend to be more entry-level in nature. From early 2023 onward, the high-exposure series drops sharply, falling below 40 per cent by late 2025, while the low-exposure series exhibits only a modest decline. The divergence is visually striking and temporally aligned with the diffusion of generative AI.

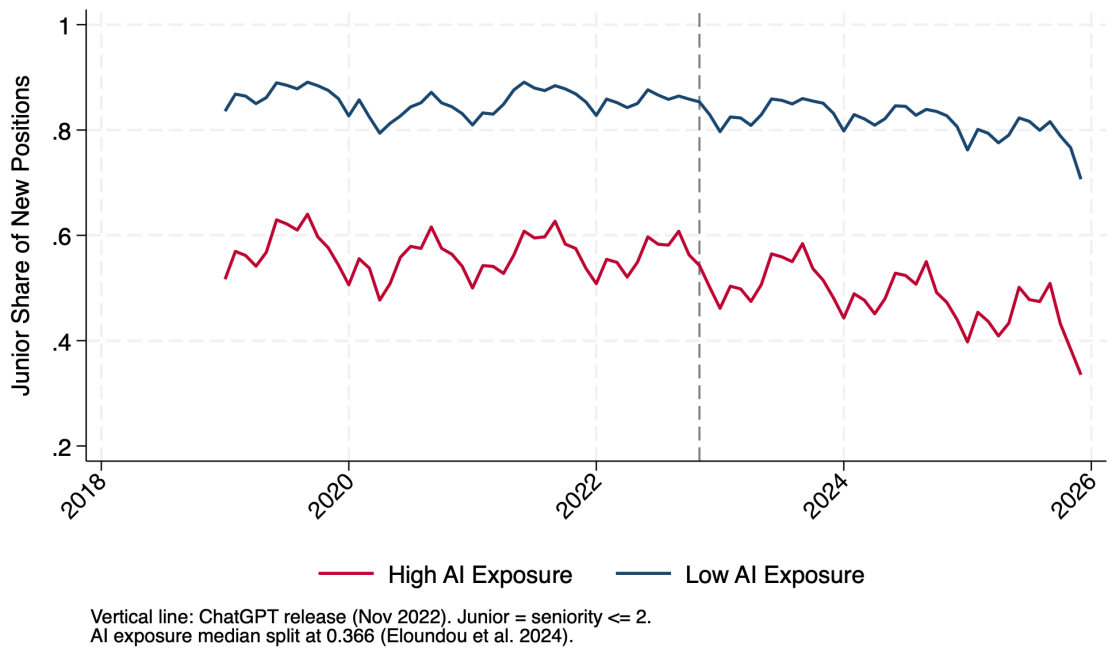

Figure 2: Monthly junior share of new position starts by AI exposure group. Junior defined as seniority ≤ 2 (Revelio Labs classification). AI exposure median split at 0.366 (Eloundou et al. (2024)). Vertical dashed line marks ChatGPT release (November 2022).

To quantify this divergence, I estimate an occupation-level difference-in-differences specification:

$$\text{JuniorShare}_{o,t} = \alpha_o + \gamma_t + \beta(\text{AI}_o \times \text{Post}_t) + \varepsilon_{o,t} \tag{9}$$

where $\text{JuniorShare}_{o,t}$ is the share of new positions with seniority level 2 or below in occupation o and half-year t, AI_o is the continuous AI exposure score, and Post_t equals one from 2023H1 onward. Observations are weighted by cell size; standard errors are clustered at the O*NET occupation level (917 clusters).

	Coefficient	**Std. error**	t	**95% CI**
$\text{AI}_o \times \text{Post}_t$	-0.096	0.011	-8.71	$[-0.117, -0.074]$

Table 2: Pooled difference-in-differences estimate. Dependent variable: junior hiring share at the occupation–half-year level, weighted by cell size. Occupation and half-year fixed effects absorbed. Standard errors clustered at the O*NET code level (917 clusters). Estimation window: 2019H1–2025H2. $N = 12,760$.

The estimated interaction coefficient is $\hat{\beta} = -0.096$ ($t = -8.71$, $p < 0.001$): after the diffusion of generative AI, occupations with higher exposure experience a disproportionately larger decline in the share of new positions filled by junior workers. A hypothetical

shift from zero to full exposure corresponds to a 9.6 percentage-point reduction in the junior share.

The pooled estimate, however, assumes that pre-treatment trends in the junior share were parallel across exposure levels. I examine this assumption through a semi-annual event study:

$$\text{JuniorShare}_{o,t} = \alpha_o + \gamma_t + \sum_{k \neq 0} \beta_k (\text{AI}_o \times \mathbb{1}[t = k]) + \varepsilon_{o,t} \tag{10}$$

where α_o and γ_t denote occupation and half-year fixed effects, respectively, and the summation runs over all half-year periods with 2022H2 ($k = 0$) as the omitted baseline. Although ChatGPT was released on 30 November 2022, I designate 2022H2 rather than 2022H1 as the reference period for two reasons. First, Lichtinger & Hosseini Maasoum (2025) document that firm-level hiring adjustments did not commence until the first quarter of 2023, implying that the final months of 2022 remained behaviourally pre-treatment. Second, the inherent lag between hiring decisions and position start dates — encompassing job posting, interviewing, and onboarding — further delays any compositional response in the Revelio data. The coefficient at 2022H1 (-0.005, $p = 0.44$) is statistically indistinguishable from the baseline, providing internal validation of this choice. Observations are weighted by cell size.

Figure 3 and Table 6 present the results. The pre-period coefficients (2019H1 to 2022H1) fluctuate within a narrow band around zero, with no monotonic trend. A mild positive hump is visible during 2019H2–2020H2, plausibly reflecting differential COVID-era hiring in high-AI-exposure occupations, which are disproportionately white-collar and remote-compatible. These fluctuations are transient and subside before the treatment window: the coefficient at 2022H1 is -0.005, statistically indistinguishable from the baseline. A formal test for a linear pre-trend, implemented by replacing the unrestricted pre-period interactions with a single occupation-specific trend term following Freyaldenhoven et al. (2019), yields an estimated slope of essentially zero, confirming the absence of a systematic pre-existing divergence.

The post-period tells a sharply different story. From 2023H1 onward, the estimates decline steeply: −0.064 at 2023H1, −0.103 at 2024H1, and −0.115 by 2025H2. All post-period coefficients are significant at the 1 per cent level, with 95 per cent confidence intervals well below zero. The decline is an order of magnitude larger than the pre-period fluctuations and — unlike the pre-period hump — is sustained over three years. A supplementary quarterly event study (see Appendix Figure 6) shows a similar timing pattern around the transition from 2022Q4 to 2023Q1, aligning with the onset documented by Lichtinger & Hosseini Maasoum (2025).

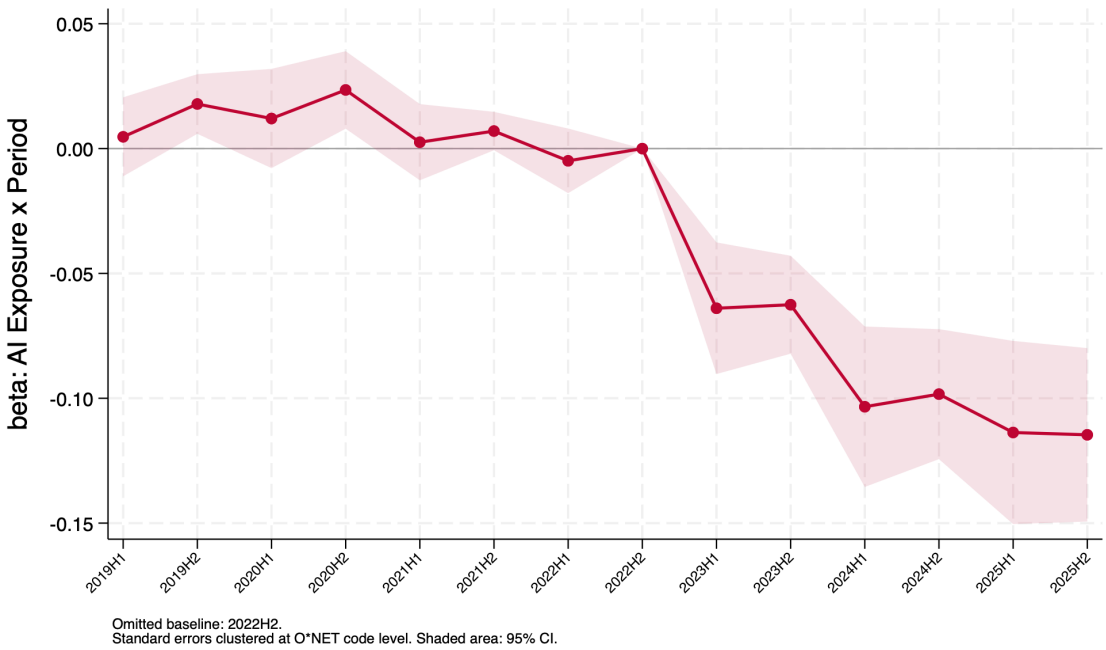

Figure 3: Event study: effect of AI exposure on junior share of new position starts. Each point represents the estimated coefficient $\hat{\beta}_k$ from Equation 10. Estimation window: 2019H1–2025H2. Baseline period: 2022H2. Standard errors clustered at the O*NET code level; shaded area shows 95% confidence interval.

In summary, high-AI-exposure occupations experienced a sustained and statistically significant decline in the junior share of new hires following the diffusion of generative AI. By 2025H2, the estimated effect reaches approximately 11.5 percentage points for a fully exposed occupation. The magnitude of the post-period break, its monotonic trajectory, and its precise temporal alignment with the diffusion of generative AI collectively support a causal interpretation, notwithstanding the minor pre-period

fluctuations. This compositional shift raises a natural follow-up question: what happens to the junior workers who would otherwise have entered these occupations?

4.1.1 Robustness Check

I assess the sensitivity of the compositional shift results to alternative specifications, with full details reported in Appendix B. Replacing the primary AI exposure measure with the AI Occupational Exposure index of Felten et al. (2021) yields a standardised coefficient of -0.018 ($t = -5.33$), close to the baseline estimate of -0.021. The result also holds under a binary treatment specification: high-exposure occupations experience a 3.9 percentage-point larger decline in the junior share relative to low-exposure occupations. I also report the original 2018H1–2025H2 event study as a window robustness check; the post-period pattern is nearly unchanged, but I keep the 2019H1–2025H2 window as the main specification because it aligns with the stated sample construction. The AI Applicability Score of Tomlinson et al. (2025), which captures observed Copilot usage rather than theoretical task exposure, produces a smaller and statistically insignificant estimate. This is not unexpected. The identification strategy relies on predetermined theoretical exposure interacted with the ChatGPT shock, and a usage-based measure introduces additional variation from adoption frictions — differences in how quickly firms actually deploy AI tools — that attenuates the estimated relationship.

Varying the seniority threshold reveals that the effect is concentrated at the junior–senior boundary. The coefficient is statistically insignificant when junior status is restricted to seniority level 1 alone, strongest at the baseline cutoff of ≤ 2, and attenuated when expanded to ≤ 3. This pattern suggests that seniority level 2 workers — those who perform substantive but routine cognitive tasks without the expertise needed to pivot toward complementary functions — constitute the primary margin of displacement.

4.2 Career Mobility

In this section, I turn to the analysis of career mobility, with particular emphasis on the post-ChatGPT period. These estimates are not direct tests of the static production model, which has no search, matching, or skill-accumulation structure. Instead, they describe worker-level adjustment margins that are empirically important if the formal compositional shift is to be understood as a labour market process. The individual-level panel structure of the Revelio Labs data allows me to track career transitions directly. This addresses a limitation explicitly acknowledged by both Lichtinger & Hosseini Maasoum (2025) and Klein Teeselink (2025), who identify the analysis of individual career trajectories following AI exposure as a direction for future research.

For observed transitions, I construct outcome variables capturing the change in seniority level, the log wage change ($\Delta \ln w$), and an indicator for whether the worker switched O*NET occupation code. For both the extensive and intensive analyses, the core specification is a triple-difference model:

$$Y_i = \beta_1 \text{Junior}_i + \beta_2 \text{Junior}_i \times \text{AI}_o + \beta_3 \text{Junior}_i \times \text{Post}_i + \beta_4 \text{AI}_o \times \text{Post}_i + \beta_5 \text{Junior}_i \times \text{AI}_o \times \text{Post}_i + \alpha_o + \gamma_{m(i)} + \varepsilon_i \tag{11}$$

where Y_i denotes the relevant mobility outcome for origin-position observation i, and α_o and $\gamma_{m(i)}$ denote occupation and reference-month fixed effects, respectively. The observation-level post indicator follows the timing definitions in Equation 7 and Equation 8. For intensive-margin outcomes, $m(i)$ is the transition month. For the extensive-margin transition indicator, $m(i)$ is the origin-position start month. The main effect of AI_o is absorbed by occupation fixed effects; the specification reports the lower-order junior interactions and the two treatment interactions of interest.

4.2.1 Extensive Margin: Who Moves?

Table 3 reports results from the extensive-margin analysis. Using the full sample of 27.5 million positions (including those without a subsequent transition), I estimate the triple-difference specification with an indicator for whether a transition is observed as the dependent variable. The estimate of β_4 is 0.548, indicating that senior workers

in high-exposure occupations became substantially *more* mobile after ChatGPT. This pattern is compatible with an informal extension of the model in which higher marginal products raise outside options and poaching, but the formal model itself does not contain this mobility channel. It could also reflect senior workers leaving deteriorating matches in exposed occupations. I therefore interpret this coefficient as evidence of increased gross senior reallocation, not as a direct measure of net senior employment growth. The triple-interaction term $\beta_5 = -0.660$ is negative and highly significant: relative to senior workers, junior workers in the same high-exposure occupations experienced a substantially *smaller* increase in transition probability. The net effect for junior workers ($\beta_4 + \beta_5 \approx -0.11$) is negative, implying that juniors in high-AI-exposure occupations became less mobile after ChatGPT while their senior counterparts became more so. This seniority asymmetry is consistent with the model's formal junior-displacement implication, although the senior-side mechanism remains ambiguous without net flow evidence. The result is robust to excluding positions that began after July 2024 ($\beta_5 = -0.662$), suggesting that the censoring of recently started positions does not drive the estimate. This test does not eliminate all right-censoring concerns, since older positions that remain ongoing at the data endpoint are still coded as non-transitions. The extensive-margin estimates should therefore be interpreted as transition probabilities observed within the current panel window rather than completed career-spell outcomes.

	Full Sample	Excluding Recent
Junior	–0.075***	–0.044***
	(0.010)	(0.010)
Junior × AI	0.205***	0.151***
	(0.019)	(0.018)
Junior × Post	0.377***	0.370***
	(0.009)	(0.008)
AI × Post	0.548***	0.567***
	(0.028)	(0.028)
Junior × AI × Post	–0.660***	–0.662***
	(0.031)	(0.029)
N	27,471,965	24,531,422
R^2	0.181	0.158
Occupation FE	Yes	Yes
Month FE	Yes	Yes

Table 3: Career mobility regressions: extensive margin (probability of job transition)

Notes: *Dependent variable is an indicator for whether the worker transitions to a new position. Column (1) uses the full sample; column (2) excludes positions starting after 2024m6 to assess censoring of recently started positions. All specifications are linear probability models with occupation and month fixed effects. Standard errors clustered at the O*NET occupation level in parentheses.* *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

4.2.2 Intensive Margin: Transition Outcomes

Conditional on transitioning, Column (1) of Table 4 shows that junior workers leaving high-exposure occupations after ChatGPT gain significantly more seniority in their next position ($\beta_5 = 0.122$). This outcome has no direct counterpart in the formal model, which treats junior and senior workers as fixed types and contains no skill-accumulation process. It should instead be interpreted as evidence on post-shock career progression among workers who transition. The estimate is susceptible to a selection concern: if AI disproportionately displaces weaker junior workers, those who successfully transition constitute a positively selected subsample, and the observed seniority gain may overstate any genuine upskilling effect.

To address this concern, I implement a Heckman (1979) two-step correction. The selection equation models the probability of transitioning as a function of the main regressors, occupation fixed effects, and a firm-level exclusion restriction: the pre-period

mean separation rate at the individual's company, entered both as a main effect and interacted with junior status.[4] The separation rate is a strong predictor of transitions in the first stage ($z = 343$). The probit is estimated on a 10 per cent random subsample for computational tractability, and the resulting inverse Mills ratio is included in the outcome equation estimated on the full sample.[5]

Column (2) reports the corrected estimates. The estimated inverse Mills ratio is negative ($\hat{\lambda} = -0.129$), indicating that workers with higher unobserved propensity to transition tend to experience *smaller* seniority gains upon moving — a pattern of negative selection on the outcome dimension. Crucially, however, the triple-interaction point estimate is virtually unchanged after correction ($\beta_5 = 0.125$, compared with 0.122 in the baseline). Because the corrected standard errors do not include Murphy–Topel or bootstrap adjustment, the precision of Column (2) should be interpreted cautiously. The main lesson from the Heckman exercise is therefore the stability of the point estimate, rather than the exact reported standard error. The seniority gain for junior workers in high-AI-exposure occupations after ChatGPT is consistent with genuine skill-upgrading among transitioning juniors, rather than compositional upgrading of the transition pool through selective attrition. This interpretation goes beyond the static model and should be read as an empirical mechanism suggested by the worker-level data.

Columns (3) and (4) examine log wage changes upon transition, controlling for the change in seniority level to isolate wage effects net of promotion-related gains. The triple-interaction term β_5 is essentially zero and statistically insignificant (0.007, $t = 0.51$), so I do not detect a differential junior wage penalty in the Revelio predicted salary

[4]The exclusion restriction logic is that firms with historically higher turnover generate more transitions mechanically, but conditional on transitioning, the origin firm's historical separation rate should not affect the seniority change achieved in the new position. The interaction with junior status allows the relationship between firm turnover and transition probability to differ across seniority groups.

[5]The probit is estimated on observations where $\text{mod}(\text{user_id}, 10) = 0$, yielding approximately 2.7 million observations. The reported standard errors in the corrected specifications do not incorporate the Murphy–Topel adjustment or a bootstrap correction for generated-regressor uncertainty. They may therefore understate uncertainty for both the inverse Mills ratio and the second-stage treatment coefficients. I use the corrected specifications primarily to assess the stability of the point estimates relative to the baseline columns.

data. The coefficient on $\text{AI}_o \times \text{Post}_i$ is negative and significant (-0.111), suggesting that transitions originating from high-exposure occupations are associated with lower wage growth overall, but this effect does not differ by seniority. The detectable adjustment is therefore more visible in employment composition and mobility than in differential wage compression.

Columns (5) and (6) examine occupational switching. The triple-interaction term is small and insignificant ($\beta_5 = -0.008$, $t = -0.61$ in the baseline). Junior workers do not systematically reallocate toward lower-AI-exposure occupations after ChatGPT. This null result is informative: the post-ChatGPT adjustment in career mobility operates along the seniority dimension rather than through occupational flight. Workers who transition move to positions of comparable AI exposure but at higher seniority — a pattern more consistent with upward filtering within related occupations than with wholesale retreat from AI-exposed work.

	Seniority Gain		**Log Salary Gain**		**Occupation Switch**	
	(1)	(2)	(3)	(4)	(5)	(6)
Junior	1.534***	1.517***	–0.085***	–0.084***	0.059**	0.067**
	(0.049)	(0.049)	(0.014)	(0.014)	(0.027)	(0.027)
Junior × AI	–0.617***	–0.605***	0.031	0.030	–0.034	–0.040
	(0.094)	(0.094)	(0.026)	(0.026)	(0.047)	(0.047)
Junior × Post	–0.114***	–0.117***	0.032***	0.032***	–0.006	–0.004
	(0.015)	(0.015)	(0.007)	(0.007)	(0.007)	(0.007)
AI × Post	0.019	0.035	–0.111***	–0.112***	0.020	0.013
	(0.025)	(0.025)	(0.015)	(0.015)	(0.014)	(0.014)
Junior × AI × Post	0.122***	0.125***	0.007	0.006	–0.008	–0.009
	(0.030)	(0.030)	(0.013)	(0.013)	(0.013)	(0.013)
Δ Seniority			0.203***	0.203***		
			(0.002)	(0.002)		
IMR		–0.129***		0.011***		0.057***
		(0.014)		(0.003)		(0.006)
N	12,487,780	12,487,780	10,676,852	10,676,852	12,487,780	12,487,780
R^2	0.160	0.161	0.350	0.351	0.120	0.121
Occupation FE	Yes	Yes	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes	Yes	Yes
Heckman correction	No	Yes	No	Yes	No	Yes

Table 4: Career mobility regressions: intensive margin outcomes conditional on job transition

Notes: *Sample restricted to workers observed in at least two positions. Columns (1)–(2) report seniority gain (levels); columns (3)–(4) report log salary change controlling for seniority gain; columns (5)–(6) report occupation switching (linear probability model). Odd columns are baseline OLS; even columns include Heckman selection correction using firm-level pre-period separation rate as the exclusion restriction. Standard errors clustered at the O*NET occupation level in parentheses. The standard errors in the Heckman columns do not incorporate Murphy–Topel or bootstrap correction for generated-regressor uncertainty and may be understated. Significance stars in those columns should therefore be interpreted cautiously.* *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

5 Discussion

A natural question arising from the results is why I do not detect the wage movements predicted by the theoretical framework. In the model, junior employment falls because w_J falls, and senior employment rises because w_S rises. A high labour supply elasticity ε can make these wage movements small relative to the quantity response, but it does not remove the price response altogether. The absence of a detected wage effect should therefore be interpreted as a limitation of the empirical test rather than as direct confirmation of the wage predictions of the model. One important caveat is that the salary data from Revelio Labs are predicted values, imputed from job titles, locations, and company characteristics, rather than actual reported compensation. This means that the wage evidence is available, but less reliable than the employment and mobility measures. It is worth noting that Lichtinger & Hosseini Maasoum (2025), who draw on the same Revelio Labs data infrastructure, do not report any wage-based analysis, which may itself reflect an awareness of this limitation. A more precise test of the price channel is therefore an important direction for future research, especially with administrative wage records or other datasets containing reliable individual compensation.

The Heckman correction in the mobility analysis provides a further window into mechanisms beyond the formal model. The estimated inverse Mills ratio is negative ($\hat{\lambda} = -0.129$), indicating that workers with higher unobserved propensity to transition tend to experience smaller seniority gains upon moving. In the standard parameterisation, this implies negative selection on the outcome dimension: the unobservables

that make a worker more likely to transition are associated with *worse*, not better, destination matches. The key finding, however, is that the triple-interaction point estimate is virtually unchanged after correction (0.125 versus 0.122 in the baseline). Since the reported standard errors in the corrected columns do not include Murphy–Topel or bootstrap adjustment, the precision of these estimates may be overstated. I therefore place more weight on the stability of the point estimate than on the exact corrected standard error. The preferred empirical interpretation is one of genuine skill-upgrading among transitioning juniors: AI tools may help capable junior workers in exposed occupations build competencies faster, enabling larger seniority gains upon transition. The occupation-switching null reinforces this reading — junior workers in high-exposure occupations are not fleeing to unrelated fields but moving upward within the same occupational domain. Because the theoretical framework treats junior and senior workers as fixed types, this result should be viewed as a complementary career-progression finding rather than as a formal model prediction.

The extensive-margin results merit further discussion. Senior workers in high-AI-exposure occupations become substantially more mobile after ChatGPT, while the net effect for junior workers is negative. One natural concern is whether increased senior mobility should be read as evidence for the model's prediction that senior employment rises ($\partial n_S/\partial A > 0$). The answer is no: the extensive-margin estimate captures gross transitions, not inflows, outflows, or stocks. A senior transition can reflect poaching into a better match, which would be consistent with augmentation. It can also reflect exit from deteriorating conditions in exposed occupations. Distinguishing these mechanisms would require constructing occupation-level inflows, outflows, and net senior headcount changes from the position start and end dates. Such an analysis is feasible with a richer use of the Revelio Labs position data, but lies beyond the scope of the current paper. I therefore interpret senior mobility as evidence of increased reallocation among senior workers, not as a standalone confirmation of senior augmentation.

A broader concern for this literature is the lack of consensus on how to measure occupational exposure to AI. Gimbel et al. (2026) document that the most widely used indices — including those of Eloundou et al. (2024), Felten et al. (2021), and Webb (2020) — disagree substantially on which occupations are most exposed, reflecting differences in whether exposure is defined through task descriptions, technical benchmarks, or patent-based measures. This motivates my use of three methodologically distinct indices: the primary LLM exposure score based on expert and GPT-4 assessment of task-level automation potential, the AI Occupational Exposure index derived from mapping AI capabilities to occupational abilities, and the AI Applicability Score constructed from revealed usage patterns in real-world AI interactions. If the main findings hold across all three, this substantially strengthens the confidence that the results are not driven by a particular measurement choice. A separate data limitation concerns the representativeness of the Revelio Labs sample. Because the data are drawn from public LinkedIn profiles, the sample likely over-represents white-collar, professional occupations relative to the working population as a whole. Since LLM exposure is concentrated in precisely such occupations, this sample composition may amplify the estimated effects relative to what a nationally representative survey would yield.

Finally, as noted in Section 3, the seniority threshold separating junior from senior workers is held fixed throughout the analysis, yet AI may itself reshape expertise requirements within occupations, shifting this boundary endogenously over time. Modelling such a response would require the between-occupation general equilibrium framework of Autor & Thompson (2025) and is left for future work. More broadly, the estimates presented here capture the early stages of generative AI diffusion. The capabilities of the technology are expanding rapidly — from text-based assistance toward autonomous task execution — and the occupational task bundles that define current exposure scores may not remain stable as these capabilities evolve. Whether the compositional shifts documented in this essay persist, intensify, or reverse as firms and workers adapt over

a longer horizon remains an open question, and one that longer post-treatment panels will be well placed to address.

6 Conclusion

This essay has argued that aggregate null results for AI's employment effects conceal meaningful compositional shifts within occupations. Extending the expertise framework of Autor & Thompson (2025) to permit within-occupation task specialisation, I derive formal predictions for how AI capability improvements affect junior employment shares, senior marginal products, wages, and net occupation-level employment. The evidence is strongest for the model's junior compositional prediction: in occupations more exposed to large language models, the junior hiring share declines. The worker-level mobility results provide complementary evidence on adjustment margins beyond the static model. Juniors who transition achieve larger seniority gains, consistent with genuine skill-upgrading, while senior workers become substantially more mobile. These results should be interpreted as career-progression and gross-reallocation patterns rather than direct tests of formal model predictions. By contrast, I do not detect the wage movements predicted by the formal model in the Revelio predicted salary data, so the price-margin results should be interpreted cautiously.

These findings carry a subtler implication. Much of the public debate frames AI as a threat to entire occupations or skill groups. The evidence here suggests a different pattern: AI reshapes *who does the work within an occupation* before it reshapes *whether the occupation exists*. If this pattern generalises, then the workers most immediately affected are not those in "threatened occupations" but those at the bottom of the ladder within occupations that are, on the whole, thriving. This is a form of displacement that aggregate statistics are poorly equipped to detect, and that conventional policy responses — retraining workers for different occupations — are poorly designed to

address. The more relevant question may be how firms and institutions restructure entry-level pathways when the tasks that once served as training grounds are increasingly performed by machines.

Whether these early compositional shifts prove transitory or permanent, and whether the expertise threshold itself adapts in response, are questions that longer post-treatment data will help resolve. But if displacement does begin at the junior level, as the evidence here suggests, the implications extend beyond the immediately affected workers. Every senior professional was once a junior one; entry-level positions have traditionally served as the channel through which new labour market entrants acquire occupation-specific skills and progress toward independent judgment. A sustained contraction of this channel would warrant attention from policymakers and educational institutions, not because entire occupations are disappearing, but because the pathway into them is narrowing.

References

Acemoglu, D., & Restrepo, P. (2019). Automation and New Tasks: How Technology Displaces and Reinstates Labor. *Journal of Economic Perspectives*, *33*(2), 3–30. https://doi.org/10.1257/jep.33.2.3

Autor, D. H., & Dorn, D. (2013). The Growth of Low-Skill Service Jobs and the Polarization of the US Labor Market. *American Economic Review*, *103*(5), 1553–1597. https://doi.org/10.1257/aer.103.5.1553

Autor, D., & Thompson, N. (2025). *Expertise* (Working Paper No. 33941). https://doi.org/10.3386/w33941

Brynjolfsson, E., Chandar, B., & Chen, R. (2025). *Canaries in the Coal Mine? Six Facts about the Recent Employment Effects of Artificial Intelligence*. https://digitaleconomy.stanford.edu/publication/canaries-in-the-coal-mine-six-facts-about-the-recent-employment-effects-of-artificial-intelligence/

Eloundou, T., Manning, S., Mishkin, P., & Rock, D. (2024). GPTs are GPTs: Labor market impact potential of LLMs. *Science*, *384*(6702), 1306–1308. https://doi.org/10.1126/science.adj0998

Felten, E., Raj, M., & Seamans, R. (2021). Occupational, Industry, and Geographic Exposure to Artificial Intelligence: A Novel Dataset and Its Potential Uses. *Strategic Management Journal*, *42*(12), 2195–2217.

Freyaldenhoven, S., Hansen, C., & Shapiro, J. M. (2019). Pre-event Trends in the Panel Event-Study Design. *American Economic Review*, *109*(9), 3307–3338. https://doi.org/10.1257/aer.20180609

Garicano, L., & Rossi-Hansberg, E. (2006). Organization and Inequality in a Knowledge Economy. *The Quarterly Journal of Economics*, *121*(4), 1383–1435. https://doi.org/10.1162/qjec.121.4.1383

Gimbel, M., Kendall, J., & Kulsakdinun, R. (2026). *Labor Market AI Exposure: What Do We Know?* [Technical report]. https://budgetlab.yale.edu/research/labor-market-ai-exposure-what-do-we-know

Heckman, J. J. (1979). Sample Selection Bias as a Specification Error. *Econometrica*, *47*(1), 153–161. http://www.jstor.org/stable/1912352

Klein Teeselink, B. (2025, September 22). *Generative AI and Labor Market Outcomes: Evidence from the United Kingdom*. https://ssrn.com/abstract=5516798%20or%20 http://dx.doi.org/10.2139/ssrn.5516798

Lichtinger, G., & Hosseini Maasoum, S. M. (2025,). *Generative ai as seniority-biased technological change: evidence from u. S. Résumé and job posting data*. https://doi.org/10.2139/ssrn.5425555

Massenkoff, M., & McCrory, P. (2026, March 5). *Labor market impacts of AI: A new measure and early evidence*. https://www.anthropic.com/research/labor-market-impacts

Tomlinson, K., Jaffe, S., Wang, W., Counts, S., & Suri, S. (2025,). *Working with AI: Measuring the Occupational Implications of Generative AI*. https://arxiv.org/abs/2507.07935

Webb, M. (2020). *The Impact of Artificial Intelligence on the Labor Market*. https://doi.org/10.2139/ssrn.3482150

Appendix

A Derivation of Theoretical Model

A.1 Firm's Optimisation Problem and First-Order Conditions

The representative firm in the occupation takes output price p and factor prices w_J, w_S, r as given, and chooses AI capital M, junior labour in task L (n_J^L), junior labour in task G (n_J^G), and senior labour n_S to maximise profit:

$$\max_{M, n_J^L, n_J^G, n_S} \quad p \cdot Y - w_J\left(n_J^L + n_J^G\right) - w_S n_S - rM \tag{12}$$

where output Y is the CES aggregate defined in Equation 1, and the three task production functions are given by Equation 2.

Substituting the CES structure, the marginal revenue product of task $k \in \{L, H, G\}$ is:

$$p \cdot \frac{\partial Y}{\partial y_k} = p \cdot \beta_k \left(\frac{Y}{y_k}\right)^{\frac{1}{\sigma}} \tag{13}$$

The first-order conditions for an interior solution are:

$$p \cdot \beta_L \left(\frac{Y}{y_L}\right)^{\frac{1}{\sigma}} \cdot A = r \tag{14}$$

$$p \cdot \beta_L \left(\frac{Y}{y_L}\right)^{\frac{1}{\sigma}} \cdot a_L = w_J \tag{15}$$

$$p \cdot \beta_G \left(\frac{Y}{y_G}\right)^{\frac{1}{\sigma}} \cdot c = w_J \tag{16}$$

$$p \cdot \beta_H \left(\frac{Y}{y_H}\right)^{\frac{1}{\sigma}} \cdot b_H = w_S \tag{17}$$

Derivation of the no-arbitrage condition. Dividing Equation 14 by Equation 15:

$$\frac{A}{a_L} = \frac{r}{w_J} \quad \Rightarrow \quad w_J = \frac{a_L \cdot r}{A} \tag{18}$$

Since AI and junior labour are perfect substitutes in task L, the firm employs both only when their unit costs are equalised. This condition pins junior wages entirely as a function of AI capability A and the rental rate r, independently of conditions in tasks H and G.

Derivation of the constant task output ratio. Dividing Equation 15 by Equation 16:

$$\frac{\beta_L (Y/y_L)^{\frac{1}{\sigma}} \cdot a_L}{\beta_G (Y/y_G)^{\frac{1}{\sigma}} \cdot c} = 1 \tag{19}$$

Simplifying:

$$\frac{\beta_L a_L}{\beta_G c} \cdot \left(\frac{y_G}{y_L}\right)^{\frac{1}{\sigma}} = 1 \quad \Rightarrow \quad \frac{y_G}{y_L} = \left(\frac{\beta_G c}{\beta_L a_L}\right)^{\sigma} \equiv \gamma \tag{20}$$

The ratio γ is a structural constant determined by technology and task share parameters. Intuitively, both tasks L and G are produced using the same marginal factor (junior labour at wage w_J), so their marginal costs move simultaneously. The CES demand system then fixes their output ratio regardless of A.

Senior wage expression. From Equation 17 directly:

$$w_S = p \cdot \beta_H \left(\frac{Y}{y_H}\right)^{\frac{1}{\sigma}} \cdot b_H \tag{21}$$

This is the marginal revenue product of senior labour in task H.

A.2 Senior Allocation to Task H

The main text writes task G as a junior-assigned generic task and therefore abstracts from senior production of G. This is a simplifying modelling restriction, not a separate empirical claim. It can be interpreted as the corner solution of a more general model in which senior workers could perform generic support tasks but face a higher opportunity cost of time.

If a senior worker were reallocated to task G, the return would be no higher than the marginal revenue product of junior workers in task G, which by Equation 16 is exactly w_J. The return to senior labour in task H is w_S. Senior allocation to task H is therefore consistent with optimisation whenever $w_S > w_J$, that is:

$$p \cdot \beta_H \left(\frac{Y}{y_H}\right)^{\frac{1}{\sigma}} \cdot b_H > \frac{a_L \cdot r}{A} \tag{22}$$

This condition holds when task H is sufficiently valued relative to routine and generic support tasks. This is empirically plausible: in the Revelio Labs data, salaries for senior positions (seniority ≥ 5) substantially exceed those for entry-level positions (seniority ≤ 2) within the same O*NET occupation. I maintain $w_S > w_J$ as a regularity condition throughout.

A.3 Comparative Statics

I derive the effects of an increase in AI capability A on the equilibrium allocation in four steps.

Step 1: Junior wage and employment

From the no-arbitrage condition:

$$\frac{\partial w_J}{\partial A} = -\frac{a_L r}{A^2} < 0 \tag{23}$$

Substituting into the supply function (Equation 3):

$$n_J = \varphi_J \left(\frac{a_L r}{A}\right)^{\varepsilon} \quad \Rightarrow \quad \frac{\partial n_J}{\partial A} = -\varepsilon \varphi_J (a_L r)^{\varepsilon} A^{-(\varepsilon+1)} < 0 \tag{24}$$

An increase in AI capability monotonically reduces both the wage and employment of junior workers in the occupation. The magnitude of the employment response is governed by the supply elasticity ε: when ε is large, even a small decline in w_J induces substantial exit. Pinning down the precise value of ε is beyond the scope of this essay; the key qualitative prediction — that junior employment falls with AI capability — holds for any $\varepsilon > 0$.

Step 2: Task output levels

To characterize how task outputs respond to A, I reduce the model's dimensionality. Since $y_G = \gamma y_L$ with γ constant, I define $\tilde{\beta}_L \equiv \beta_L + \beta_G \gamma^{\frac{\sigma-1}{\sigma}}$ and write the CES aggregate as a function of two effective task outputs:

$$Y = \left[\tilde{\beta}_L y_L^{\frac{\sigma-1}{\sigma}} + \beta_H y_H^{\frac{\sigma-1}{\sigma}}\right]^{\frac{\sigma}{\sigma-1}} \tag{25}$$

From Equation 14, the shadow price of task L output satisfies:

$$p \cdot \beta_L \left(\frac{Y}{y_L}\right)^{\frac{1}{\sigma}} = \frac{r}{A} \tag{26}$$

This implicitly defines the demand for y_L as a function of y_H and A. Holding y_H fixed, an increase in A reduces the effective marginal cost of task L output (r/A), which by CES demand raises y_L. Since $y_G = \gamma y_L$, task G output rises proportionally.

To verify that total output Y rises: from Equation 25, Y is increasing in both y_L and y_H. Since y_L increases and y_H is (initially) unchanged, Y unambiguously increases.

Step 3: Senior augmentation

I establish that the demand curve for senior labour shifts outward when A increases, and that the equilibrium senior wage and employment both rise.

Define the inverse demand for senior labour, holding A fixed:

$$w_S^D(n_S; A) \equiv p \cdot \beta_H \left(\frac{Y(n_S, A)}{b_H n_S}\right)^{\frac{1}{\sigma}} \cdot b_H \tag{27}$$

where $Y(n_S, A)$ is total output evaluated at the optimal task allocation for given n_S and A.

Claim 1: w_S^D is decreasing in n_S. An increase in n_S raises $y_H = b_H n_S$ and hence Y. However, y_H rises proportionally with n_S, while Y exhibits diminishing returns to y_H under the CES structure ($\sigma < \infty$). Consequently, $Y/(b_H n_S)$ falls, and so does w_S^D.

Claim 2: w_S^D is increasing in A. Holding n_S fixed, an increase in A reduces the marginal cost of task L, raising y_L and $y_G = \gamma y_L$. Since $y_H = b_H n_S$ is unchanged, Y increases while y_H does not, so $Y/(b_H n_S)$ rises and w_S^D rises.

Equilibrium effect. The demand curve $w_S^D(n_S; A)$ is downward-sloping and shifts outward when A rises. The supply curve $n_S = \varphi_S w_S^{\varepsilon}$ is upward-sloping. Standard com-

parative statics of supply-demand intersection imply that the new equilibrium features both higher w_S and higher n_S:

$$\frac{\partial w_S}{\partial A} > 0, \quad \frac{\partial n_S}{\partial A} > 0 \tag{28}$$

The increase in n_S also raises $y_H = b_H n_S$ in equilibrium. This feedback lowers Y/y_H relative to the impact effect at fixed n_S, but it does not overturn the comparative static because the senior labour demand curve has shifted outward. This confirms the augmentation result: AI capability improvements raise senior wages and senior employment via CES task complementarity.

Step 4: Compositional shift

Define the junior employment share as $s_J \equiv n_J/(n_J + n_S)$. From Steps 1 and 3, $\partial n_J/\partial A < 0$ and $\partial n_S/\partial A > 0$. Both the numerator and denominator of s_J are affected, the junior employment share unambiguously declines.

Internal reallocation of junior labour

Although total junior employment falls, the allocation of remaining juniors between tasks L and G also shifts. From $y_G = \gamma y_L$ and $y_G = c \cdot n_J^G$:

$$n_J^G = \frac{\gamma y_L}{c} \tag{29}$$

Since y_L is increasing in A (Step 2), n_J^G increases. Combined with declining total junior employment (n_J falls), this implies:

$$n_J^L = n_J - n_J^G \tag{30}$$

which falls faster than n_J itself. Junior workers are squeezed out of task L on two margins: AI directly substitutes their labour, and the remaining juniors reallocate toward task G where AI cannot substitute.

The above derivations assume an interior solution in which $n_J^L > 0$. For sufficiently large A, all junior workers exit task L and produce only task G output. I focus on the interior case, which is empirically relevant given that junior workers remain present in high-exposure occupations throughout the sample period.

A.4 Net Occupation-Level Employment

The net effect of AI capability on total occupation-level employment $N = n_J + n_S$ is:

$$\frac{\partial N}{\partial A} = \frac{\partial n_J}{\partial A} + \frac{\partial n_S}{\partial A} \tag{31}$$

The first term is unambiguously negative and the second is unambiguously positive (from my derivation in Appendix A.3). The net sign depends on the relative magnitudes, which are governed by the structural parameters. Senior augmentation dominates ($\partial N/\partial A > 0$) when:

- The task share of high-expertise tasks (β_H) is large, amplifying the complementarity channel.
- Senior supply is highly responsive (φ_S large or ε large), so that wage increases translate into large employment gains.

Conversely, junior displacement dominates ($\partial N/\partial A < 0$) when β_L is large, junior supply is highly responsive (φ_J large or ε large), and the initial junior share is high.

B Robustness Check

B.1 Summary of Alternative Specifications

Specification	Treatment	Junior def.	Coefficient	Std. error	N
Main	z(Eloundou)	≤ 2	-0.021^{***}	(0.002)	14,587
R1: Felten	z(Felten)	≤ 2	-0.018^{***}	(0.003)	14,227
R2: Tomlinson	z(Tomlinson)	≤ 2	-0.005	(0.003)	14,227
R3: Seniority 1	z(Eloundou)	≤ 1	-0.003	(0.003)	14,587
R4: Seniority 3	z(Eloundou)	≤ 3	-0.009^{***}	(0.003)	14,587
R5: Binary	Binary(Eloundou)	≤ 2	-0.039^{***}	(0.005)	14,587

Table 5: Robustness of the pooled difference-in-differences estimate for the junior hiring share. All continuous treatment variables are standardised to have zero mean and unit standard deviation at the occupation level. The dependent variable is the mean junior share within each occupation–half-year cell, weighted by cell size. Standard errors in parentheses, clustered at the O*NET code level. $***p < 0.01$.

Notes. The baseline period is 2022H2. "z(·)" denotes the occupation-level z-score of the indicated measure. R1 uses the AI Occupational Exposure index of Felten, Raj and Seamans (2021). R2 uses the AI Applicability Score of Tomlinson et al. (2025). R5 defines high exposure as above-median Eloundou et al. (2024) score.

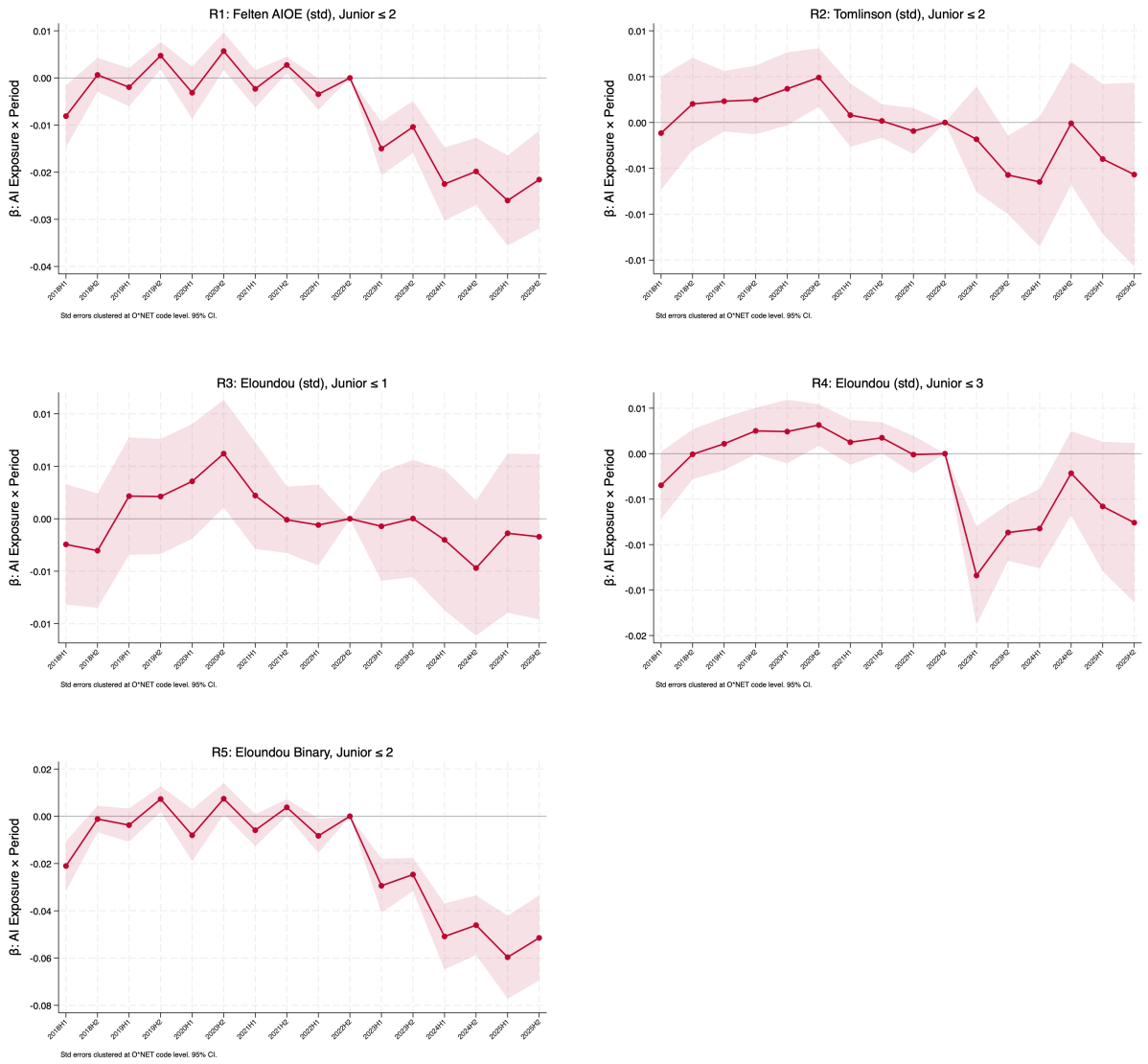

Figure 4: Robustness Check under different AI Exposure Measure and Different Junior Category Specification

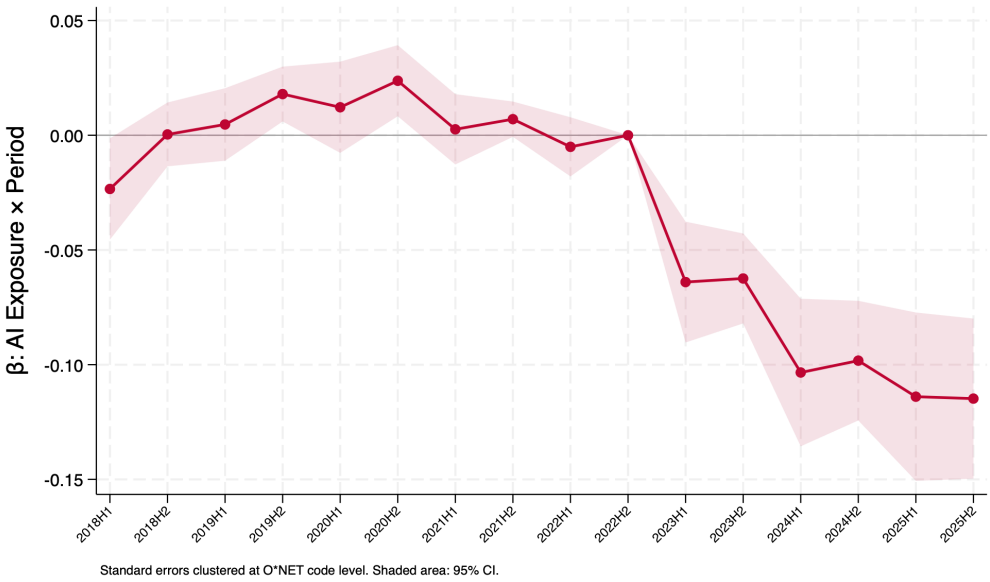

Figure 5: Extended-window event study using 2018H1–2025H2. The omitted baseline is 2022H2. This specification is reported as a robustness check because position starts before 2019 may be affected by left-censoring from the construction of the position sample.

C Other Statistics

Period	$\hat{\beta}_k$	95% CI lower	95% CI upper
2019H1	0.005	−0.011	0.020
2019H2	0.018	0.006	0.030
2020H1	0.012	−0.008	0.032
2020H2	0.024	0.008	0.039
2021H1	0.003	−0.013	0.018
2021H2	0.007	−0.001	0.015
2022H1	−0.005	−0.018	0.008
2022H2	**0**		
2023H1	−0.064	−0.090	−0.038
2023H2	−0.062	−0.082	−0.043
2024H1	−0.103	−0.136	−0.071
2024H2	−0.098	−0.124	−0.072
2025H1	−0.114	−0.151	−0.077
2025H2	−0.115	−0.150	−0.080

Table 6: Event study coefficients from Equation 10. The dependent variable is the junior hiring share at the occupation–half-year level, weighted by cell size. The estimation window is 2019H1–2025H2. The omitted baseline is 2022H2. Standard errors clustered at the O*NET code level (917 clusters).

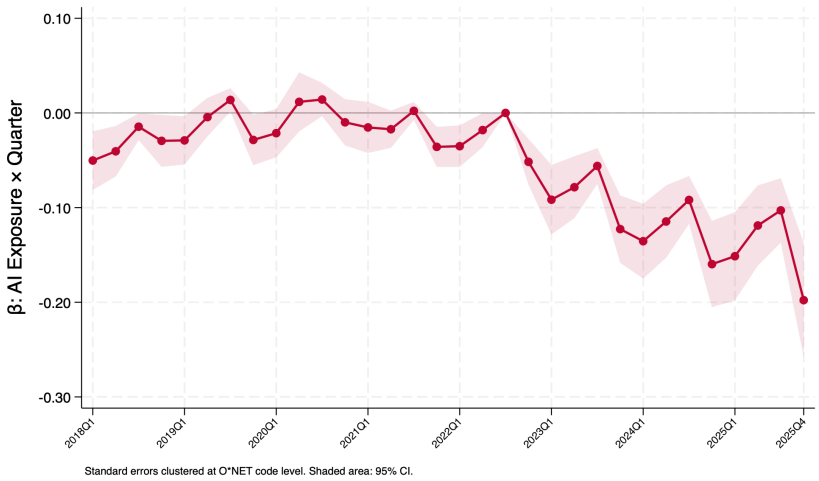

Figure 6: Supplementary quarterly event study using the extended 2018Q1–2025Q4 window. The figure is used as a timing check for the onset of the post-ChatGPT decline.